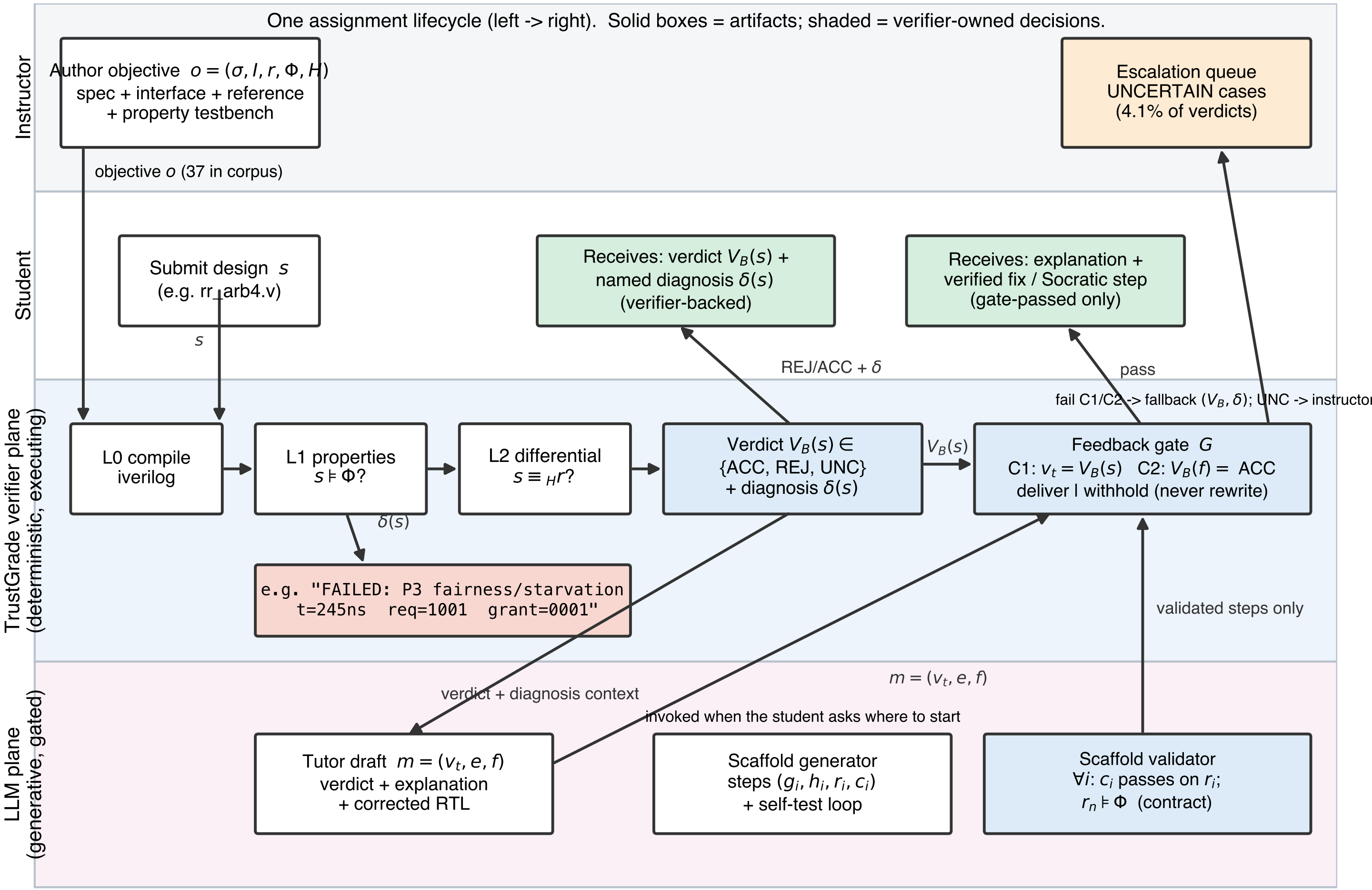

One assignment lifecycle (left -> right). Solid boxes = artifacts; shaded = verifier-owned decisions.
Instructor
Author objective $o = (\sigma, I, r, \Phi, H)$
spec + interface + reference
+ property testbench
Escalation queue
UNCERTAIN cases
(4.1% of verdicts)
objective o (37 in corpus)
Student
Submit design s
(e.g. rr_arb4.v)
s
Receives: verdict $V_B(s)$ +
named diagnosis $\delta(s)$
(verifier-backed)
Receives: explanation +
verified fix / Socratic step
(gate-passed only)
REJ/ACC + δ
pass
fail C1/C2 -> fallback (V_B, δ); UNC -> instructor
TrustGrade verifier plane
(deterministic, executing)
L0 compile
iverilog
L1 properties
$s \models \Phi$?
L2 differential
$s \equiv_H r$?
Verdict $V_B(s) \in$
{ACC, REJ, UNC}
+ diagnosis $\delta(s)$
$V_B(s)$
Feedback gate G
C1: $v_t = V_B(s)$ C2: $V_B(f) =$ ACC
deliver | withhold (never rewrite)
$\delta(s)$
e.g. "FAILED: P3 fairness/starvation
t=245ns req=1001 grant=0001"
validated steps only
LLM plane
(generative, gated)
$m = (v_t, e, f)$
verdict + diagnosis context
invoked when the student asks where to start
Tutor draft $m = (v_t, e, f)$
verdict + explanation
+ corrected RTL
Scaffold generator
steps (g_i, h_i, r_i, c_i)
+ self-test loop
Scaffold validator
$\forall i$: c_i passes on r_i;
$r_n \models \Phi$ (contract)